Tahmid Hasan Sakib

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PROFESSIONAL SUMMARY

Graduate Research Assistant evaluating Jetson-based autonomous vehicles with embedded Linux, camera/LiDAR sensing, and secure ROS 2. Validate physical perception and control behavior on distinct QCar1 and QCar2 platforms.

TECHNICAL SKILLS

Software and platforms: Python, C/C++, Linux, Git, NVIDIA Jetson TX2 and AGX Orin, ROS 2/SROS2 on QCar2, Arduino/AVR; UART, I2C, SPI, and GPIO interface testing.

Perception and integration: CSI and Intel RealSense cameras, RPLiDAR A2/2D LiDAR, YOLOv5n/YOLOv8, OpenCV, camera-LiDAR fusion, physical test validation.

EXPERIENCE

Graduate Research Assistant | Tennessee Technological University, Cookeville, TN | May 2024-Present

- Validated non-ROS QCar1 Jetson TX2 camera/LiDAR perception and three-stop navigation in clean and printed-sign physical runs.
- Tracked detector and runtime settings across 18 pre- and 18 post-mitigation attack runs and analyzed vehicle-level missed stops.
- Mapped SROS2 credential provisioning and authenticated DDS trust boundaries on QCar2; brought up selected interfaces on its AGX Orin platform.
- Compared PULSE recorded-log replay with live fallback tests and REDAP offline frames with later E3/E4 vehicle demonstrations.
- Mentored undergraduate researchers in NSF REU cohorts in summers 2025 and 2026 under faculty supervision.

Hardware Engineer Intern | Robi Axiata Limited, Dhaka, Bangladesh | December 2019-January 2020

- Prototyped AVR/Arduino systems and simulated hardware and communication behavior with MATLAB, LTspice, and Cisco Packet Tracer.

SELECTED ENGINEERING PROJECTS

QCar1 physical navigation | Jetson TX2, CSI camera, RPLiDAR A2, YOLOv5n

Camera detections and LiDAR ranging supported a three-stop, two-turn physical route without preloaded stop-sign waypoints.

QCar2 secure ROS 2 evaluation | AGX Orin, ROS 2 Humble, SROS2

Controlled middleware proof of concept connected authenticated-message manipulation with observed qualitative vehicle behavior.

PULSE physical fallback | QCar2 live runs

Across 10 black/white fallback-on runs, 18/20 patched-sign encounters were flagged and all 20 stop sequences completed; stop-line position was not independently measured.

Five-stage RISC processor | Verilog, Icarus Verilog, GTKWave

Implemented and simulated a 32-bit IF/ID/EX/MEM/WB design; exercised arithmetic, load/store, and branch testbenches.

EDUCATION

Tennessee Technological University | M.S., Electrical and Computer Engineering, expected December 2026 | GPA 4.00/4.00

Master's thesis defended; official degree conferral pending.

Military Institute of Science and Technology, Dhaka, Bangladesh | B.Sc., Computer Science and Engineering, August 2021

SELECTED PUBLICATIONS

First author, QCar1 adversarial-perception study, IEEE Access, accepted for publication, 2026; first author, secure ROS 2 supply-chain study, IEEE MILCOM 2025 workshop.